

ESTAS-II — Transport Framework Reference Manual

ESTAS-AQUABC Development Team

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1 Introduction

ESTAS-II (Ecological-System Transport and Aggregated Simulation, version II) is a one-dimensional box-model transport framework written in Fortran 90/95. It provides the hydrodynamic “scaffolding” — advection, dispersion, settling, mass loads, withdrawals, and sediment fluxes — around the AQUABC ecological model, enabling multi-box simulations of lakes, reservoirs, estuaries, and coastal waters.

ESTAS-II can operate standalone (reading all forcing from text files) or be embedded as a library inside larger hydrodynamic codes such as SHYFEM.

1.1 Key Features

- Arbitrary number of well-mixed boxes connected by advective and dispersive links
- Forward-Euler mass-balance solver with sub-daily time steps
- Supports three sediment coupling modes: off, prescribed fluxes, or full diagenesis
- Allelopathic interactions among phytoplankton groups
- Macroalgae and pelagic exergy diagnostics
- Resuspension modelling with critical shear stress thresholds
- Text and binary output formats; optional COCOA project-style outputs
- Command-line configurability for input file, constant overrides, binary output, and shear stress table

2 Program Architecture

2.1 Execution Flow

The main program `ESTAS_II` performs the following steps:

1. **Parse command-line arguments** — determine input file, optional constant file, binary output file, and shear-stress file.
2. **Read configuration** — `READ_AQUATIC_MODEL_INPUTS` loads `INPUT.txt` and all referenced sub-files (bathymetry, forcing time series, initial conditions, etc.).
3. **Run simulation** — `RUN_SIMULATION` executes the main time loop.
4. **Clean up** — deallocate pelagic and sediment arrays, close output units.

2.2 Module Organisation

Module / File	Purpose
<code>ESTAS_II.f90</code>	Main program: argument parsing, init, run, cleanup
<code>mod_GLOBAL.f90</code>	Global constants, dimensions (<code>nkn</code> , <code>nstate</code> = 32, sediment dims), allocatable arrays for state variables, derivatives, driving functions, flags, process rates, sediment variables
<code>mod_AQUATIC_MODEL.f90</code>	<code>AQUATIC_MODEL_DS</code> derived type aggregating all sub-model data; <code>READ_AQUATIC_MODEL_INPUTS</code> master reader
<code>mod_SIMULATE.f90</code>	<code>RUN_SIMULATION</code> — outer repeat loop, inner time-step loop, output writing, cost-function evaluation
<code>mod_SOLVER.f90</code>	<code>PELAGIC_SOLVER</code> — <code>SOLVE</code> routine: time-function update, <code>CALC_DERIV</code> , Euler mass update, sediment integration
<code>mod_PELAGIC_BOX_MODEL.f90</code>	<code>PELAGIC_BOX_MODEL_DS</code> derived type: boxes, links, boundary conditions, derivative arrays
<code>mod_PELAGIC_BOX.f90</code>	<code>PELAGIC_BOX_DS</code> — per-box data: volume, areas, concentrations, masses
<code>mod_PELAGIC_LINK.f90</code>	Advective and dispersive link data structures
<code>mod_PELAGIC_ECOLOGY.f90</code>	Interface to <code>AQUABC</code> pelagic kinetics (<code>PELAGIC_KINETICS</code> call)
<code>mod_BOTTOM_SEDIMENTS.f90</code>	Sediment model initialisation: constant loading, <code>isedi</code> parameter, allocation
<code>mod_INITIAL_CONDITIONS.f90</code>	Reading initial concentrations from files
<code>mod_INITIALIZE_PELAGIC_BOX_MODEL.f90</code>	Construction of box-model topology from input data
<code>mod_INITIALIZE_AQUATIC_MODEL.f90</code>	High-level initialisation orchestrator
<code>mod_TIME_SERIES.f90</code>	<code>TIME_SERIE</code> type and interpolation for forcing data
<code>mod_INTERPOLATE.f90</code>	General interpolation utilities
<code>mod_MASS_LOAD.f90</code>	Mass-load and withdrawal handling

Module / File	Purpose
mod_RESUSPENSION.f90	Resuspension algorithms: shear stress, critical thresholds
mod_BASIN.f90	Basin geometry helpers
mod_PELAGIC_EXERGY.f90	Eco-exergy computation
mod_COST_FUNCTION.f90	Obs-vs-model cost function
mod_UTILS_01.f90	General utility routines
sub_READ_PELAGIC_INPUTS.f90	Read pelagic forcing files
sub_READ_BOTTOM_SEDS_FLUXES_INPUTS.f90	Read prescribed sediment flux files
sub_WRITE_PELAGIC_OUTPUT.f90	ASCII pelagic output writer
sub_WRITE_PELAGIC_BINARY_OUTPUT.f90	Binary pelagic output
sub_WRITE_PELAGIC_MEM_OUTPUT.f90	In-memory pelagic output
sub_WRITE_PELAGIC_MEM_BINARY_OUTPUT.f90	In-memory binary pelagic output
sub_WRITE_PELAGIC_MEM_SAVED_OUTPUT.f90	Saved-state output
sub_WRITE_PELAGIC_MEM_SAVED_BINARY_OUTPUT.f90	Saved-state binary output
sub_WRITE_PELAGIC_EX_MEM_OUTPUT.f90	Exergy output
sub_WRITE_PELAGIC_EX_MEM_BINARY_OUTPUT.f90	Binary exergy output

3 Global Dimensions and Constants

The `mod_GLOBAL` module defines the fixed dimensions of the coupled system:

Parameter	Value	Description
<code>nstate</code>	32	Pelagic state variables
<code>nconst</code>	318	Pelagic model constants
<code>n_driving_functions</code>	10	Environmental driving functions
<code>nflags</code>	5	Pelagic flag variables
<code>n_saved_outputs</code>	5	Saved pelagic outputs carried across time steps
<code>NDIAGVAR</code>	30	Pelagic diagnostic (process-rate) variables
<code>NUM_SED_VARS</code>	24	Sediment state variables
<code>NUM_SED_CONSTS</code>	171	Sediment model constants
<code>NUM_SED_DRIV</code>	1	Sediment driving functions
<code>NUM_FLUXES_TO_SEDIMENTS</code>	24	Water-column → sediment flux variables
<code>NUM_FLUXES_FROM_SEDIMENTS</code>	30	Sediment → water-column flux variables
<code>NDIAGVAR_sed</code>	25	Sediment diagnostic variables
<code>NUM_SED_OUTPUTS</code>	26	Sediment output variables
<code>NUM_SED_FLAGS</code>	3	Sediment model flags
<code>NUM_SED_SAVED_OUTPUTS</code>	5	Sediment saved outputs
<code>NUM_ALLOLOPATHY_STATE_VARS</code>	4	Allelopathic state variables (appended to pelagic)

The number of boxes (`nkn`) is determined at runtime from the box-model input file.

4 Input Configuration

4.1 Master Input File (INPUT.txt)

The simulation is driven by a sequential text file read on Fortran unit 10. Lines beginning with # are treated as comments. The file is read record-by-record in the following order:

Record	Example	Description
1–5	# DESCRIPTION ...	Header / description lines (5 lines)
6	1998	Base year
7	6209.0	Simulation start time (days since reference)
8	6574.0	Simulation end time (days)
9	1	Number of repeat cycles
10	240	Time steps per day
11	10	Print interval (in time steps)
12	INPUTS/	Pelagic model input folder
13	PELAGIC_INPUTS.txt	Pelagic model input file name
14	OUTPUTS/	Pelagic model output folder
15	2	Resuspension option (0 = off)
16	0	MODEL_BOTTOM_SEDIMENTS (0 = off, 1 = prescribed, 2 = diagenesis)
17	2	Number of prescribed sediment flux sets
18+	filenames	Prescribed sediment flux file names (one per set)

Additional records follow for sediment model input when MODEL_BOTTOM_SEDIMENTS = 2, COCOA options, and other advanced settings.

4.2 Pelagic Input Files

The pelagic input folder must contain the file named in record 13 (typically PELAGIC_INPUTS.txt), which in turn references:

- **Box-model topology** — number of boxes (**nkn**), volumes, surface and bottom areas
- **Advective links** — ADVECTIVE_LINKS.txt: upstream/downstream box pairs and flow time-series references
- **Dispersive links** — DISPERSIVE_LINKS.txt: box pairs, dispersion coefficients, interface areas
- **Flow time series** — FLOW_TS.txt and individual FORC_TS_*.txt files
- **Boundary conditions** — open-boundary concentrations per state variable
- **Initial conditions** — per-box starting concentrations (files or inline)
- **Forcing time series** — air temperature, cloud cover, evaporation, wind speed, solar radiation
- **Bathymetry** — BATHYMETRY_*.txt (one per box): depth–area–volume profiles
- **Mass loads** — tributary inflows with concentrations

- **Mass withdrawals** — outflow points

4.3 Sediment Input Files

4.3.1 Prescribed Fluxes (Mode 1)

When `MODEL_BOTTOM_SEDIMENTS = 1`, the files listed after record 17 provide time-varying fluxes $F_j(t)$ for each state variable, read by `sub_READ_BOTTOM_SEDS_FLUXES_INPUTS.f90`.

4.3.2 Full Diagenesis (Mode 2)

When `MODEL_BOTTOM_SEDIMENTS = 2`, ESTAS reads additional input:

- `BOTTOM_SEDIMENT_MODEL_INPUT.txt` — layer geometry (depths, porosities, densities), particle mixing coefficients, diffusion coefficients, burial rates, initial sediment concentrations, sediment model constants
- `EXTRA_WCONST.txt` — extra water-column constants for sediment coupling

4.4 Resuspension Input

When `RESUSPENSION_OPTION > 0`:

- `CRITICAL_SHEAR_STRESSES.txt` — per-box critical shear stress τ_c values
- Shear stress and resuspension concentration time series files

5 Transport Model

5.1 Box-Model Formulation

ESTAS represents the water body as **nkn** well-mixed boxes. Each box i has a time-varying volume $V_i(t)$, surface area A_i^s , and bottom area A_i^b . The mass of state variable j in box i is:

$$M_{i,j} = C_{i,j} \cdot V_i$$

where $C_{i,j}$ is the concentration (g m^{-3}).

5.2 Mass-Balance Equation

The total time derivative of mass in box i for state variable j is the sum of seven components:

$$\frac{dM_{i,j}}{dt} = \left(\frac{dM}{dt}\right)_{\text{adv}} + \left(\frac{dM}{dt}\right)_{\text{disp}} + \left(\frac{dM}{dt}\right)_{\text{settl}} + \left(\frac{dM}{dt}\right)_{\text{load}} + \left(\frac{dM}{dt}\right)_{\text{withdr}} + \left(\frac{dM}{dt}\right)_{\text{kin}} + \left(\frac{dM}{dt}\right)_{\text{sed}}$$

5.2.1 Advection

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{adv}} = \sum_{k \in \text{in}(i)} Q_k C_{k,j}^{\text{up}} - \sum_{k \in \text{out}(i)} Q_k C_{i,j}$$

where Q_k is the flow rate through advective link k , and $C_{k,j}^{\text{up}}$ is the upstream concentration (or boundary concentration for open boundaries).

5.2.2 Dispersion

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{disp}} = \sum_k \frac{D_k A_k^{\text{ifc}}}{L_k} (C_{\text{nbr},j} - C_{i,j})$$

where D_k is the dispersion coefficient ($\text{m}^2 \text{s}^{-1}$), A_k^{ifc} is the interface area (m^2), and L_k is a characteristic mixing length.

5.2.3 Settling

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{settl}} = -w_{s,j} A_i^b f_{\text{part},j} C_{i,j}$$

where $w_{s,j}$ is the settling velocity (m s^{-1}), A_i^b is the bottom area, and $f_{\text{part},j}$ is the particulate fraction ($1 - f_{\text{diss},j}$).

5.2.4 Mass Loads

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{load}} = \sum_{\ell} Q_{\ell} C_{\ell,j}^{\text{load}}$$

where Q_{ℓ} and $C_{\ell,j}^{\text{load}}$ are the flow and concentration of load source ℓ .

5.2.5 Mass Withdrawals

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{withdr}} = - \sum_w Q_w C_{i,j}$$

5.2.6 Kinetics (AQUABC)

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{kin}} = V_i \cdot R_{i,j}$$

where $R_{i,j}$ is the reaction rate from the AQUABC pelagic kinetics model ($\text{g m}^{-3} \text{ day}^{-1}$, converted to seconds internally).

5.2.7 Prescribed Sediment Fluxes

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{sed}} = A_i^b \cdot F_j^{\text{sed}}(t)$$

where $F_j^{\text{sed}}(t)$ is the prescribed areal flux ($\text{g m}^{-2} \text{ s}^{-1}$), active only in Mode 1.

6 Time Integration

6.1 Forward Euler Scheme

ESTAS uses a simple forward-Euler (explicit) scheme. At each time step Δt :

$$M_{i,j}^{n+1} = M_{i,j}^n + \left. \frac{dM_{i,j}}{dt} \right|^n \cdot \Delta t$$

$$V_i^{n+1} = V_i^n + \left. \frac{dV_i}{dt} \right|^n \cdot \Delta t$$

$$C_{i,j}^{n+1} = \frac{M_{i,j}^{n+1}}{V_i^{n+1}}$$

The solver ID (PELAGIC_SOLVER_NO = 1) currently selects this scheme. The infrastructure supports future addition of higher-order methods (e.g. Runge–Kutta 2).

6.2 Stability and Safety

- **Minimum concentration clamp:** if $C_{i,j} < \text{MIN_CONCENTRATION}$, it is reset to MIN_CONCENTRATION and the mass recomputed.
- **Maximum concentration clamp:** if $C_{i,j} > 10^{10}$, the simulation is halted.
- **Negative mass warning:** diagnostic messages are printed whenever $M_{i,j}^{n+1} < 0$.
- **Time step:** the user sets `TIME_STEPS_PER_DAY` in `INPUT.txt`. Typical stable values range from 48 to 480.

6.3 Simulation Control

- **Repeat cycles:** the simulation can be repeated `NUM_REPEATS` times, useful for spin-up.
- **Print interval:** output is written every `PRINT_INTERVAL` time steps.

7 Sediment Coupling Modes

ESTAS supports three modes of sediment–pelagic interaction, controlled by `MODEL_BOTTOM_SEDIMENTS`:

7.1 Mode 0 — No Sediments

No sediment fluxes are applied. Only pelagic processes contribute to the kinetic derivative.

7.2 Mode 1 — Prescribed Sediment Fluxes

Time-varying fluxes per state variable are read from external files. Each flux set provides $F_j(t)$ values that are interpolated to the current time and applied as:

$$\left(\frac{dM_{i,j}}{dt}\right)_{\text{sed}} = A_i^b \cdot F_j^{\text{sed}}(t)$$

Multiple flux sets can be active simultaneously (e.g. sandy-sediment fluxes and muddy-sediment fluxes in different spatial regions).

7.3 Mode 2 — Full Diagenesis (AQUABC Sediment Model)

The AQUABC sediment diagenesis model is called at each time step for each box. This mode:

1. Passes settling fluxes from the water column as upper boundary conditions
2. Solves the multi-layer sediment equations (24 state variables $\times$ `NUM_SED_LAYERS` layers)
3. Returns dissolved fluxes back to the water column (30 flux variables)

The returned fluxes replace the prescribed sediment flux term in the mass balance. See the *AQUABC Reference Manual* for full details of the diagenesis equations, bioturbation, and bioirrigation.

7.3.1 Sediment Arrays

In Mode 2, the following 3-D arrays are allocated per box, layer, and variable:

- `INIT_SED_STATE_VARS(nkn, NUM_SED_LAYERS, NUM_SED_VARS)` — initial conditions
- `FINAL_SED_STATE_VARS(nkn, NUM_SED_LAYERS, NUM_SED_VARS)` — end-of-step values
- `FLUXES_FROM_SEDIMENTS(nkn, NUM_FLUXES_FROM_SEDIMENTS)` — 30 return fluxes
- `FLUXES_TO_SEDIMENTS(nkn, NUM_FLUXES_TO_SEDIMENTS)` — 24 settling deposit fluxes
- `SED_OUTPUTS(nkn, NUM_SED_LAYERS, NUM_SED_OUTPUTS)` — diagnostic outputs

8 Settling and Deposition

8.1 Settling Velocity

Each state variable has a settling velocity $w_{s,j}$ and a dissolved fraction $f_{\text{diss},j}$. The effective particulate settling flux from box i is:

$$J_{i,j}^{\text{settl}} = w_{s,j} (1 - f_{\text{diss},j}) C_{i,j}$$

8.2 Deposition to Sediments

The fraction of settled material that deposits on the bottom (as opposed to being resuspended or being laterally transported) is controlled by the *effective deposition fraction* $f_{\text{dep},j}$ and the *deposition area ratio* $r_{\text{dep},j}$:

$$F_{i,j}^{\text{dep}} = J_{i,j}^{\text{settl}} \cdot f_{\text{dep},j} \cdot r_{\text{dep},j} \cdot A_i^b$$

This deposited flux is what enters the sediment model (Mode 2) or is simply removed from the water column (Modes 0 and 1).

9 Resuspension

When `RESUSPENSION_OPTION > 0`, ESTAS applies bed-sediment resuspension to selected boxes. The module `mod_RESUSPENSION.f90` implements:

9.1 Shear-Stress Threshold

Resuspension occurs when the bed shear stress τ_b exceeds the critical shear stress τ_c for a box:

$$E = \begin{cases} E_0 \left(\frac{\tau_b}{\tau_c} - 1 \right) & \text{if } \tau_b > \tau_c \\ 0 & \text{otherwise} \end{cases}$$

where E_0 is the erosion rate coefficient.

9.2 Configuration

- `CRITICAL_SHEAR_STRESSES.txt` provides per-box τ_c values.
- Shear stress time series are read from forcing files.
- `SHUT_DOWN_SETTLING` flag optionally disables settling during resuspension events.

10 Allelopathy

ESTAS carries four additional state variables (`NUM_ALLOLOPATHY_STATE_VARS = 4`) appended after the 32 standard pelagic variables. These represent allelopathic substance concentrations released by competing phytoplankton groups.

The allelopathic module (`mod_ALLELOPATHY.f90`) computes:

- Production rates as functions of phytoplankton biomass
- Inhibition effects on susceptible phytoplankton growth
- Decay of allelopathic substances

These variables are transported by the same advection–dispersion–settling equations as the standard state variables.

11 Macroalgae

The macroalgae module (`mod_MACROALGAE.f90`) models attached benthic macroalgae using a Droop cell-quota framework. Macroalgae interact with the pelagic model through:

- Nutrient uptake from the water column (N, P)
- Oxygen production / consumption
- Light competition (shading)

Macroalgae biomass is not transported but is associated with the bottom of specific boxes.

12 Exergy Diagnostics

The module `mod_PELAGIC_EXERGY.f90` computes eco-exergy — a thermodynamic measure of ecosystem organisation. For each box, exergy components are computed from biomass concentrations using weighting factors (beta values) that reflect the genetic information content of each organismal group:

$$\text{Ex}_i = \sum_j \beta_j C_{i,j} V_i$$

Exergy outputs can be written to separate output files when enabled.

13 Output Files

13.1 Pelagic Output

ESTAS produces time-series output for each box:

Column	Content
1	Time (days)
2	Box number
3–34	Concentrations of 32 pelagic state variables
35–38	Allelopathic state variables (if active)

Output can be in ASCII or binary format, controlled by command-line arguments.

13.2 Sediment Output (Mode 2)

When full diagenesis is active:

- **Sediment concentrations** — per-box, per-layer, per-variable (file: box-specific or COCOA format)
- **Sediment fluxes to water column** — 30 flux variables per box per output step
- **Sediment burial rates** — per-box, per-variable (optional COCOA output)
- **Fluxes to sediments** — settling fluxes entering the sediment surface

13.3 COCOA Outputs

Optional extended output format (enabled via `PRODUCE_COCOA_OUTPUTS`), producing additional files for:

- Pelagic process rates
- Sediment process rates
- Burial rates
- Bidirectional sediment fluxes

14 Building and Running

14.1 Compilation

The project uses a top-level Makefile:

```
# Clean and build in release mode
make clean-lib
make build-estas BUILD_TYPE=release

# Debug build
make build-estas BUILD_TYPE=debug
```

The build discovers all `.f90` files in `SOURCE_CODE/` and compiles them via `make_lib.sh`, producing the `ESTAS_II` executable.

14.2 Running a Simulation

```
# Default: reads INPUT.txt from current directory
./ESTAS_II

# Specify input file
./ESTAS_II INPUT_gf_release.txt

# With constant override file
./ESTAS_II INPUT.txt WCONST_02.txt

# With constant override + binary output
./ESTAS_II INPUT.txt WCONST_02.txt output.bin

# Full: input + constants + binary + shear stress
./ESTAS_II INPUT.txt WCONST_02.txt output.bin CRITICAL_SHEAR_STRESSES.txt
```

14.3 Running Tests

```
# Fortran unit tests (bioturbation, etc.)
make test-fortran

# Python/pytest integration tests (if configured)
pytest tests/
```

15 Source File Reference

15.1 ESTAS Framework Files

All files are located under SOURCE_CODE/ESTAS/:

File	Lines	Description
ESTAS_II.f90	151	Main program
mod_GLOBAL.f90	286	Global dimensions, allocatable arrays, switches
mod_AQUATIC_MODEL.f90	—	Master data structure and input reader
mod_SIMULATE.f90	832	Time loop, output orchestration
mod_SOLVER.f90	1590	Derivative calculation and Euler update
mod_PELAGIC_BOX_MODEL.f90	—	Box-model derived type
mod_PELAGIC_BOX.f90	—	Per-box data structure
mod_PELAGIC_LINK.f90	—	Link data structures
mod_PELAGIC_ECOLOGY.f90	—	AQUABC kinetics interface
mod_BOTTOM_SEDIMENTS.f90	—	Sediment initialisation
mod_INITIAL_CONDITIONS.f90	—	Initial concentration reader
mod_INITIALIZE_PELAGIC_BOX_MODEL.f90	—	Topology builder
mod_INITIALIZE_AQUATIC_MODEL.f90	—	High-level init
mod_TIME_SERIES.f90	—	Time-series type and interpolation
mod_INTERPOLATE.f90	—	Interpolation utilities
mod_MASS_LOAD.f90	—	Mass loads and withdrawals
mod_RESUSPENSION.f90	—	Resuspension algorithms
mod_BASIN.f90	—	Basin geometry
mod_PELAGIC_EXERGY.f90	—	Eco-exergy computation
mod_COST_FUNCTION.f90	—	Cost function
mod_UTILS_01.f90	—	General utilities

15.2 AQUABC Ecological Model Files

Located under SOURCE_CODE/AQUABC/:

Directory	Key Files	Description
.	mod_AQUABC_II_GLOBAL.f90	AQUABC global constants
PELAGIC/	aquabc_II_pelagic_kinetics.f90	90-variable pelagic kinetics
PELAGIC/	aquabc_II_wc_outputs.f90	Water-column output mapping
SEDIMENTS/	aquabc_II_sediment_model_1_fast.f90	Multi-layer diagenesis driver
SEDIMENTS/	aquabc_II_sediment_bioturbation.f90	Biodegradation and bioirrigation
SEDIMENTS/	aquabc_II_sediment_model_constants.f90	Sediment constants
SEDIMENTS/AQUABC_SEDIMENT_LIBRARY/	library files	DOC mineralisation, redox, pH correction, etc.

15.3 Auxiliary Modules

Directory	Key Files	Description
SOURCE_CODE/ALLELOPATHY/	mod_ALLELOPATHY.f90	Allelopathic interactions
SOURCE_CODE/MACROALGAE/	mod_MACROALGAE.f90	Macroalgae (Droop quota)
SOURCE_CODE/CO2SYS/	CO2SYS source files	Carbonate equilibrium

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